# MEHMET DEMIRCI

## AI SOLUTIONS ENGINEER

Turkey, UAE & remotely available worldwide
hello@mdemirci.dev · LinkedIn profile

### PROFESSIONAL PROFILE

AI Solutions Engineer with 7+ years of experience designing, developing and deploying production machine-learning systems. Specializes in LLM agents, retrieval pipelines, document intelligence and computer vision, with a track record of integrating reliable AI capabilities into enterprise, transport and industrial environments.

### EXPERIENCE

#### AI Solutions Engineer

innoscripta SE · Germany · Hybrid | Oct 2021 – Sep 2026

- Led discovery and delivery for enterprise document-intelligence products, translating operational workflows into scoped extraction services and measurable acceptance criteria.
- Delivered invoice, email and timesheet parsers at 90%+ field-level accuracy and integrated them into existing operational workflows.
- Designed agentic data-extraction pipelines for European enterprise information, including entity resolution, knowledge-graph integration and controls for reviewable AI decisions.
- Created stable FastAPI service contracts and AWS-based delivery pipelines so product teams could adopt AI capabilities without depending on model internals.
- Advised delivery teams on model trade-offs, integration constraints, failure modes and production-readiness decisions.

#### ML Solutions Engineer · Applied AI Delivery

ISSD Bilişim Elektronik A.Ş. · Turkey · Hybrid | Aug 2018 – Sep 2021

- Delivered real-time incident-detection systems for highways and tunnels with Intel and fortiss, aligning model behaviour with operational response needs and roadside hardware limits.
- Contributed to a 3D container-scanning solution for an Istanbul port, evaluating new sensing hardware and improving operational throughput while reducing misreads.
- Supported a Huawei rail-safety engagement in Serbia, defining detection around the time available for operators to act rather than model accuracy in isolation.
- Designed, developed and deployed roadside variable-message-sign systems across multiple Turkish cities, covering the full path from engineering decisions to field commissioning.

# PRODUCTS & SELECTED PROJECTS

## INDEPENDENT PRODUCTS

### WorkGuard - Worker Safety Monitoring

Independent product in development | 2026

Developing a camera-based worker-safety product, starting with person, safety-helmet and high-visibility clothing detection. Fine-tuned YOLO26s and ran training and inference locally on Apple Silicon. Built a reproducible video pipeline with annotated output and per-frame detection records. Selected public-footage demonstration; human review required. Equipment-to-person association and reliable compliance alerts remain development work.

Watch the prototype demonstration

## PROFESSIONAL PROJECTS BY COMPANY

innoscripta SE · Enterprise AI

### Document Intelligence & Process Automation

Led the delivery of a shared extraction service for invoices, emails and timesheets. Scoped outputs around the fields downstream teams actually needed, established stable API contracts and moved three tools into production at 90%+ field-level accuracy. Since launch, the service has saved more than €60,000 in manual processing costs.

innoscripta SE · Data and AI architecture

### Enterprise Knowledge Graph

Designed the extraction approach for fragmented European enterprise data, combining agentic workflows, entity resolution, graph storage and semantic search. Focused the architecture on traceability and controlled failure modes so automated conclusions could be reviewed and corrected.

ISSD · Intel, fortiss and Huawei partnerships

### Safety-Critical Vision Systems

Supported solution design and field delivery for tunnel, highway and rail-crossing detection systems. Balanced latency, false-alert rates and edge-hardware constraints, helping move computer-vision systems from model evaluation into live transport environments.

## TECHNICAL FOUNDATION

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|---|---|
| Applied AI | LLM agents, RAG, NLP, document AI, object detection, time-series forecasting |
| Architecture | FastAPI, asynchronous services, microservices, knowledge graphs, vector search |
| Infrastructure | AWS, Docker, Kubernetes, CI/CD, OpenVINO edge compute |
| Engineering | Python, C++, Qt, PostgreSQL, JanusGraph, AWS Neptune, Qdrant, Elasticsearch, Redis, MongoDB |

## EDUCATION

MSc, Sustainable Environment and Energy Systems
Middle East Technical University, Northern Cyprus Campus · 2017

BSc, Mechatronics & Automation Systems Engineering
Eastern Mediterranean University · 2014–2017

## LANGUAGES

Turkish · Native
English · Fluent
German · Beginner

## SELECTED PUBLICATIONS

Mehmet Demirci · Incident Detection on Junction Using Image Processing (2021) · Application of the Neural Network Dependability Kit in Real-World Environments (2020) · Aiming for Smart Wind Energy (2019)